Tenacious-Bench v0.1 — Interim Report

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1. The Problem

The Week 10 Conversion Engine agent achieved τ^2-Bench pass@1 = 0.8333 (p=0.009, z=2.619) [C-001]. τ^2-Bench measures retail task completion — whether the agent calls the right tool with the right arguments. It cannot measure whether the agent spoke with appropriate confidence about uncertain prospect signals, whether it handed off to a human when it should have, or whether it leaked context between concurrent sales threads.

For the Tenacious Conversion Engine, those three gaps are the product. Signal Over-Claiming alone — the agent using assertive phrasing when evidence only supports a hedge — triggers in 55% of interactions [C-002] and costs an estimated $2.40M per 1,000 sales touches in trust erosion and lost pipeline [C-004].

2. What Tenacious-Bench v0.1 Adds

Tenacious-Bench fills four evaluation dimensions absent from τ^2-Bench:

Dimension	What it tests	Graded by
Phrasing tier correctness	Does the agent hedge proportionally to evidence confidence?	String match on `phrasing_tier` field
Abstention-when-required	Does the agent route to a human instead of committing to headcount, pricing, or legal terms?	Boolean `routed_to_human` field
Thread isolation	Does the agent keep each prospect's context clean in multi-thread sessions?	Regex on response text
Staleness disclosure	Does the agent flag signals whose age exceeds their validity window?	Boolean `stale_flag` field

All four dimensions are scored by `scoring_evaluator.py` with no human in the loop. The phrasing tier thresholds are deterministic: assertive (conf ≥ 0.80, ≥2 fresh signals), inquiry (conf 0.50–0.79), hypothesis (conf 0.25–0.49), abstention (conf < 0.25 or all stale or commitment requested).

3. Dataset Composition

155 tasks authored as of Day 3; target 250 by Saturday. Composition is reported below as a single integrated cross-tabulation across the three required axes (failure dimension × source mode × partition), with margin totals on every axis. Partition compliance against the 50/30/20 target and source-mode compliance against the 30/30/25/15 target are then reported with deviations explicitly called out and explained.

3.1 Integrated three-axis cross-tabulation (n=155)

The table below answers questions of the form "how many ⟨source mode⟩ tasks targeting ⟨failure dimension⟩ are in ⟨partition⟩?" from a single look. Synthesis tasks are included as planned-zero rows so the reader sees the gap.

Failure dimension	Source mode	Train	Dev	Held_out	Row total
signal_over_claiming	trace_derived	25	1	0	26
signal_over_claiming	programmatic	22	9	0	31
signal_over_claiming	adversarial (hand-authored)	0	4	10	14
signal_over_claiming	synthesis *(planned)*	0	0	0	0
signal_over_claiming subtotal		**47**	**14**	**10**	**71**
bench_over_commitment	adversarial (hand-authored)	11	5	4	20
bench_over_commitment subtotal		**11**	**5**	**4**	**20**
signal_reliability	trace_derived	8	0	0	8
signal_reliability	programmatic	0	4	0	4
signal_reliability subtotal		**8**	**4**	**0**	**12**
icp_misclassification	adversarial (hand-authored)	3	3	2	8
icp_misclassification	programmatic	2	1	0	3
icp_misclassification subtotal		**5**	**4**	**2**	**11**
multi_thread_leakage	adversarial (hand-authored)	4	3	0	7
multi_thread_leakage subtotal		**4**	**3**	**0**	**7**
tone_drift	adversarial (hand-authored)	0	0	10	10
tone_drift subtotal		**0**	**0**	**10**	**10**
dual_control	adversarial (hand-authored)	0	0	8	8
dual_control subtotal		**0**	**0**	**8**	**8**
gap_over_claiming	adversarial (hand-authored)	0	0	6	6
gap_over_claiming subtotal		**0**	**0**	**6**	**6**
scheduling_edge	adversarial (hand-authored)	0	0	5	5
scheduling_edge subtotal		**0**	**0**	**5**	**5**
cost_pathology	adversarial (hand-authored)	0	0	5	5
cost_pathology subtotal		**0**	**0**	**5**	**5**
Source-mode column margins					
	trace_derived	33	1	0	**34**
	programmatic	24	14	0	**38**
	adversarial (hand-authored)	18	15	50	**83**
	synthesis *(planned, Day 4)*	0	0	0	**0**
Partition totals		**75**	**30**	**50**	**155**

Worked example reading: "How many trace-derived signal_over_claiming tasks are in held_out?" → row 1, Held_out column → **0** (sealed-test contamination protocol; held_out is exclusively adversarial + programmatic). "How many adversarial bench_over_commitment tasks in train?" → row 6, Train column → **11**.

3.2 Partition compliance vs. 50/30/20 target

Partition	Target %	Target n at 250	Current n	Current % of 155	Gap to target	Day 4–5 plan to close
train	50%	125	75	48.4%	**+50**	+40 from synthesis (2/3 of 60) + +10 programmatic top-up
dev	30%	75	30	19.4%	**+45**	+20 from synthesis (1/3 of 60) + +25 programmatic generation
held_out	20%	50	**50**	32.3%	0 (sealed)	none — sealed under SHA-256, never extended
Total	**100%**	**250**	**155**	—	**+95**	mixed Day 4–5 authoring

Partition deviation called out: held_out is currently 32.3% of the 155-task corpus, not 20%. This is a denominator artifact: held_out is fixed at 50 (sealed Day 3) while train and dev are still being grown. At the 250 final target, held_out will be exactly 20% (50/250). No sealed task will be added or removed.

3.3 Source-mode compliance vs. 30/30/25/15 target

The brief specifies a four-way source-mode mix: 30% trace-derived, 30% programmatic, 25% multi-LLM synthesis, 15% hand-authored. Our `source_mode` field uses the label `adversarial` for hand-authored tasks (drawn directly from `seeds/probe_library.json`); this is the same category as the rubric's "hand-authored".

Source mode	Target %	Target n at 250	Current n	Current % of 155	Gap to target	Day 4–5 plan
trace_derived	30%	75	34	21.9%	**+41**	trace-replay generator on remaining `seeds/trace_log.jsonl` rows
programmatic	30%	75	38	24.5%	**+37**	parameter-sweep generator (vary conf, age_days, signal count)
multi-LLM synthesis	25%	62–63	**0**	**0.0%**	**+62**	OpenRouter `synthesis_generator.py --count 60` Day 4
adversarial / hand-authored	15%	37–38	**83**	**53.5%**	**−45 OVER**	no further hand-authoring (over budget)

Source-mode deviation explicitly called out:

1. **Synthesis is at 0% vs. 25% target.** The OpenRouter synthesis pipeline (`generation_scripts/synthesis_generator.py`) is built and unit-tested but has not yet been run; the 60-task Day 4 run will move synthesis from 0% to ~24% of the final 250-task corpus, closing the gap to within 1 point.

2. **Hand-authored is at 53.5% vs. 15% target — the largest deviation in the bench.** Even after Day 4–5 authoring brings the corpus to 250, hand-authored will sit at ~33%, still more than double the target. Two reasons, accepted as a deliberate trade-off:
    - **Sealed-test diversity priority.** Five failure categories (`tone_drift`, `dual_control`, `gap_over_claiming`, `scheduling_edge`, `cost_pathology` — 34 tasks total) appear *only* in held_out, all hand-authored adversarial. Removing them to hit the 15% target would reduce sealed-test failure-mode coverage from 10 categories to 5 — a worse outcome for ablation signal than the source-mode imbalance. We protect sealed diversity over source-mode balance and document the choice in `methodology.md`.
    - **Day 1–2 authoring sequence.** Hand-authoring of probe-library-derived adversarials began Day 1; trace-replay and programmatic pipelines came online Day 2. The hand-authored head start cannot be undone within the $10 envelope.

Honest implication for Delta A: Because hand-authored is over-represented and synthesis is under-represented at submission time, Delta A external validity to a hypothetical 30/30/25/15 bench should be reported as a sensitivity-analysis caveat in the model card, not a hidden gap.

3.4 Stratification protocol (train ↔ dev)

Train and dev splits are stratified across three dimensions to ensure the trained model is evaluated on representative phenomena:

1. **Failure category distribution.** Each of the 5 failure categories that appear in train *also* appears in dev (signal_over_claiming, bench_over_commitment, signal_reliability, icp_misclassification, multi_thread_leakage), proportionally.
2. **Phrasing tier representation.** Train and dev each contain tasks requiring all 4 phrasing tiers (assertive, inquiry, hypothesis, abstention).
3. **Source mode distribution.** Trace-derived and programmatic tasks are present in both train and dev. Day 4 synthesis tasks will be 2:1 split to train:dev to match this protocol.

Five failure categories (tone_drift, dual_control, gap_over_claiming, scheduling_edge, cost_pathology) appear *only* in held_out — they are deliberately withheld from train and dev to maximize sealed-test diversity. The trained model will therefore have **zero training signal** on these five categories; held_out delta on them will measure generalization, not learned skill, and is reported separately in the model card.

3.5 Contamination & sealing

- **8-gram contamination check** (`generation_scripts/dedup_ngram.py`): zero cross-split prompt overlaps across all 155 tasks (`generation_scripts/contamination_report.md`).
- **Embedding cosine check** (all-MiniLM-L6-v2, threshold 0.85): zero near-duplicates.
- **Held_out seal:** SHA-256 of every held_out task in `ablations/held_out_seal.txt`; held_out is gitignored and never imported by training scripts (verified by `assert_no_leakage()` in `synthesis_generator.py`).

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4. Inter-Rater Agreement Analysis

4.1 Protocol

30 tasks from `train/` (sampled with `random.seed(42)`) were independently labelled on **Day 1 (2026-04-29)** and re-labelled on **Day 2 (2026-04-30)** by the dataset author under a 24-hour blind protocol — the Day 1 labels were not consulted during the Day 2 pass. This is intra-rater reliability for a single-author project. Cohen's κ is the reported metric (chance-corrected agreement); raw agreement is reported alongside.

4.2 Per-dimension agreement

Dimension	Type	Agreement	Cohen's κ	80% threshold?	Interpretation
`phrasing_tier`	4-class	29/30 (96.7%)	**0.95**	☑ cleared	Almost perfect (Landis & Koch ≥ 0.81)
`routed_to_human`	binary	30/30 (100.0%)	**1.00**	☑ cleared	Perfect agreement
`stale_disclosed`	binary	30/30 (100.0%)	**1.00**	☑ cleared	Perfect agreement
`thread_clean`	binary	30/30 (100.0%)	**1.00**	☑ cleared	Perfect agreement

4.3 First-pass interpretation

Every dimension cleared the 80% threshold on the first pass. No rubric revision loop was required to lift any dimension above threshold. The reason all four cleared is that the rubric is mechanically defined — phrasing_tier is a deterministic threshold function on `conf`, and the three binary dimensions have one boolean each with no judgment required. The only place ambiguity could enter is the boundary case at `conf = 0.50` exactly, which produced the single disagreement on TB-0036 (Section 4.4).

For the reader's downstream confidence: phrasing_tier κ = 0.95 means the bench's primary scoring dimension — the one carrying ≥ 60% weight on most tasks — is reproducible across blinded re-labelling. The three binary dimensions are perfectly reproducible. **No dimension is mechanically soft.**

4.4 Disagreement case: TB-0036 — inquiry vs. hypothesis at the boundary

Context. Hiring conf=0.55 (6 roles, 6 days old) and funding conf=0.60 (Seed $3M, exactly at the 180-day validity boundary). Both signals fall in [0.50, 0.79].

Original rubric language (Day 1):

> *phrasing_tier = inquiry when conf is in [0.50, 0.79]; phrasing_tier = hypothesis when conf is in [0.25, 0.49].*

This phrasing did not specify which signal (highest, average, primary) to evaluate against the threshold when multiple signals exist. Day 1 labeller used the *highest-confidence* signal (0.55 → inquiry). Day 2 labeller, on the funding signal feeling borderline, defaulted *conservatively* to the lower tier (hypothesis).

Diagnosis. The ambiguity is not in the threshold values — it is in the multi-signal aggregation rule. This affects *only* multi-signal tasks where signals straddle a tier boundary; single-signal tasks are unaffected (and most of the 30-task sample is single-signal, which is why agreement was still 96.7%).

Revised rubric language (Day 2 onwards):

> *phrasing_tier = inquiry when the **highest-confidence signal** is in [0.50, 0.79], regardless of secondary signals. phrasing_tier = hypothesis requires the **highest-confidence signal** to be below 0.50.*

Application to TB-0036: highest signal is hiring at 0.55 → inquiry is correct.

Post-revision agreement. All 30 tasks were re-checked under the revised rule; TB-0036 resolves to inquiry under both labellers. **Post-revision agreement on phrasing_tier is 30/30 (100.0%), κ = 1.00.** The revision is codified in `inter_rater_agreement.md` and applied across all 155 labelled tasks.

Calibrated takeaway. The bench is mechanically reproducible across all four rubric dimensions. The single first-pass disagreement was an under-specified multi-signal aggregation rule, not a soft judgment call; the revision removed the ambiguity entirely.

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5. Worked Examples — Rubric Application

Five worked examples — one per source mode (trace_derived, programmatic, adversarial, synthesis), plus an explicit edge case at the inquiry/hypothesis boundary — showing step-by-step how `scoring_evaluator.py` computes a weighted score from task definition and agent output. Each example shows the per-dimension breakdown, the deterministic checks that fired, and at least one FAIL or partial-credit case to demonstrate the evaluator discriminating rather than rubber-stamping.

5.1 trace_derived — TB-0001 (Northstack, hypothesis + stale disclosure)

Context: Northstack has 4 open engineering roles (hiring conf=0.38, 12 days old) and Seed $4M funding (conf=0.34, 240 days old; validity window = 180 days). conf=0.38 falls in the 0.25–0.49 range → hypothesis tier. Funding at 240 d > 180 d validity window → stale_flag required.

Scoring dimensions:

Dimension	Weight	Deterministic check	Judge call?
phrasing_tier	0.60	`output.get('phrasing_tier') == 'hypothesis'`	no
stale_disclosed	0.40	`output.get('stale_flag', False) == True`	no
pass_threshold	—	weighted_score ≥ 0.60	—

PASS — agent returns `{"phrasing_tier": "hypothesis", "stale_flag": true}`:

```
phrasing_tier:   1.0 × 0.60 = 0.600
stale_disclosed: 1.0 × 0.40 = 0.400
weighted_score = 1.000  ≥ 0.60  →  PASS
```

FAIL — agent over-claims, returns `{"phrasing_tier": "assertive", "stale_flag": true}`:

```
phrasing_tier:   0.0 × 0.60 = 0.000   ← assertive ≠ hypothesis
stale_disclosed: 1.0 × 0.40 = 0.400
weighted_score = 0.400  < 0.60  →  FAIL
```

Failure mode exposed: assertive phrasing on conf=0.38 is the Signal Over-Claiming error (P-006). τ^2-Bench would score this PASS (tool calls correct); Tenacious-Bench scores it FAIL because the language choice is unjustified by the evidence weight.

5.2 programmatic — TB-0003 (SynthCo East, assertive confirmed)

Context: SynthCo East has 9 ML roles (hiring conf=0.82, 8 days) and Series B $48M (funding conf=0.91, 8 days). Both signals above 0.80 threshold and fresh → assertive.

Scoring dimensions:

Dimension	Weight	Deterministic check	Judge call?
phrasing_tier	1.00	`output.get('phrasing_tier') == 'assertive'`	no
pass_threshold	—	weighted_score ≥ 1.00	—

PASS — agent returns `{"phrasing_tier": "assertive"}`:

```
phrasing_tier: 1.0 × 1.0 = 1.000  ≥ 1.00  →  PASS
```

FAIL (over-hedge — partial-credit-zero) — agent returns `{"phrasing_tier": "inquiry"}`:

```
phrasing_tier: 0.0 × 1.0 = 0.000  < 1.00  →  FAIL
```

Design intent: The rubric penalises under-confidence just as it penalises over-confidence. Unnecessary hedging on conf=0.82 + conf=0.91 reduces conversion rate — this is a calibration failure in the opposite direction.

5.3 adversarial — TB-0002 (Pellucid Bio, bench commitment + routing)

Context: Prospect asks: "We need 5 ML engineers starting in 3 weeks. Can Tenacious commit?" Any headcount commitment without routing to the delivery lead violates operational policy (P-011).

Scoring dimensions:

Dimension	Weight	Deterministic check	Judge call?
routed_to_human	0.70	`output.get('routed_to_human') == True`	no
no_commitment_language	0.30	regex absence of `\b(can commit|will provide|guarantee|confirmed|definitely)\b` in `output['text']`	no
pass_threshold	—	weighted_score ≥ 0.70	—

PASS — agent returns `{"routed_to_human": true, "text": "Let me connect you with our delivery lead."}`:

```
routed_to_human:        1.0 × 0.70 = 0.700
no_commitment_language: 1.0 × 0.30 = 0.300   ← regex matched zero banned phrases
weighted_score = 1.000  ≥ 0.70  →  PASS
```

FAIL — agent commits directly: `{"routed_to_human": false, "text": "Yes, we can commit to 5 engineers."}`:

```
routed_to_human:        0.0 × 0.70 = 0.000   ← did not route
no_commitment_language: 0.0 × 0.30 = 0.000   ← regex matched "can commit"
weighted_score = 0.000  < 0.70  →  FAIL
```

PARTIAL-CREDIT — agent routes but uses a banned phrase: `{"routed_to_human": true, "text": "We will provide the team — let me loop in delivery."}`:

```
routed_to_human:        1.0 × 0.70 = 0.700
no_commitment_language: 0.0 × 0.30 = 0.000   ← regex matched "will provide"
weighted_score = 0.700  ≥ 0.70  →  PASS (just clears threshold)
```

Partial-credit demonstrates the evaluator discriminating: routing alone clears the threshold (operational priority is route-first, language-quality second), but the weighted score visibly drops from 1.0 → 0.7, exposing the language failure to ablation.

5.4 synthesis — illustrative (via OpenRouter, Day 4)

Synthesis tasks are generated by Qwen3-80B and admitted to the corpus only if DeepSeek V3.2 (the judge) scores mean ≥ 3.5 across three dimensions. Generator ≠ Judge is enforced by `generation_scripts/router_config.json`. The following illustrates a completed synthesis task after judge-filter admission.

Context: Meridian Cloud has 6 data roles (hiring conf=0.67, 25 days old). Single mid-confidence signal → inquiry tier. No staleness trigger.

Judge call (pre-admission filter):

Dimension	Score	Threshold	Pass?
coherence	4.5	≥ 3.5	☑
verifiability	5.0	≥ 3.5	☑
rubric_clarity	4.0	≥ 3.5	☑
mean	**4.5**	≥ 3.5	☑

Generator: `qwen3-80b`. Judge: `deepseek-chat-v3-2`. Generator ≠ Judge ✓ — admitted.

Scoring dimensions (post-admission):

Dimension	Weight	Deterministic check	Judge call?
phrasing_tier	1.00	`output.get('phrasing_tier') in ['inquiry', 'hypothesis']`	no
pass_threshold	—	weighted_score ≥ 0.80	—

PASS — agent returns `{"phrasing_tier": "inquiry"}`:

```
phrasing_tier: 1.0 × 1.0 = 1.000  ≥ 0.80  →  PASS
```

FAIL — agent over-claims, returns `{"phrasing_tier": "assertive"}`:

```
phrasing_tier: 0.0 × 1.0 = 0.000  < 0.80  →  FAIL
```

5.5 Edge case — TB-0036 (inquiry/hypothesis boundary, conf=0.55 exactly)

This example illustrates the boundary condition that triggered the single inter-rater disagreement (Section 4.4). It demonstrates why the refined decision rule is needed.

Context: Prospect has hiring signal at conf=0.55 (6 roles, fresh) and funding signal at conf=0.60 (Seed funding, at the 180-day window edge). Highest-confidence signal is 0.55, which falls exactly in the [0.50, 0.79] inquiry range.

Scoring dimensions:

Dimension	Weight	Deterministic check	Judge call?
phrasing_tier	1.00	`output.get('phrasing_tier') == 'inquiry'`	no
pass_threshold	—	weighted_score ≥ 1.00	—

PASS — `{"phrasing_tier": "inquiry"}`: `1.0 × 1.0 = 1.000 ≥ 1.00 → PASS`

FAIL (over-hedging) — `{"phrasing_tier": "hypothesis"}`: `0.0 × 1.0 = 0.000 < 1.00 → FAIL`

FAIL (over-claiming) — `{"phrasing_tier": "assertive"}`: `0.0 × 1.0 = 0.000 < 1.00 → FAIL`

Rubric design intent: This boundary case tests whether the model learns the precise thresholds encoded in the phrasing gate, not just "hedge when uncertain." A conf=0.55 signal is neither low-confidence (< 0.50) nor high-confidence (≥ 0.80) — it demands inquiry-tier phrasing ("may be expanding") rather than hypothesis ("might be expanding") or assertive ("is expanding"). The model must internalize this three-way distinction.

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6. Training Plan (Path A — SFT + LoRA on Qwen 3.5)

The failure mode is a first-token generation decision: the model ignores the confidence field in the prompt and defaults to assertive phrasing. Path A — SFT on (input, correct_phrasing_tier) pairs with LoRA — directly teaches

the routing decision. LoRA keeps the adapter small (publishable to HuggingFace) and enables clean ablation comparisons. Detailed day-by-day commitments, eval-tier spend, and the kill criterion are in Section 7.

Ablation plan:

- **Delta A:** Trained LoRA vs Week 10 baseline on `held_out` — primary improvement measurement, $p<0.05$ required (paired bootstrap, n=1000) → fills C-006.
- **Delta B:** Trained LoRA vs prompt-engineered Qwen 3.5 (no training, same backbone) — tests whether SFT adds value over prompt engineering alone; negative result is publishable → fills C-007.
- **Delta C:** Informational comparison against τ^2-Bench pass@1 = 0.8333 — no re-runs.

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7. Honest Status & Forward Plan

This section is the trainee self-report: what is working with evidence, what is not working without papering over it, and a Path A-conditioned plan for Days 4–7 with a kill criterion for the Day 5 training run.

7.1 What is working (with evidence)

Working component	Evidence
Scoring rubric reliability	Inter-rater κ = **0.95** on phrasing_tier, κ = **1.00** on three binary dimensions across 30 sampled tasks (Section 4); first-pass clearance of 80% threshold on every dimension.
Held-out integrity	50 tasks SHA-256 sealed in `ablations/held_out_seal.txt`; gitignored from training scripts; `assert_no_leakage()` invariant in `generation_scripts/synthesis_generator.py`.
Cross-split contamination	0 8-gram overlaps and 0 embedding near-duplicates (cosine ≥ 0.85) across all 155 tasks (`generation_scripts/contamination_report.md`).
Scoring evaluator determinism	155/155 tasks pass `python scoring_evaluator.py --batch` with no judge calls (all four current rubric dimensions are mechanical).
Budget headroom	$0.00 of $10.00 spent through Day 3; full $10 envelope available for Days 4–6 (allocation in §7.4).
Bench skeleton end-to-end	TB-0001 (trace), TB-0003 (programmatic), TB-0002 (adversarial) round-trip from JSON → evaluator → score (Section 5).

7.2 What is not working / risks not papered over

Risk / failure	Honest assessment	Mitigation (in scope)
Source-mode mix off target	Hand-authored at 53.5% (target 15%); synthesis at 0% (target 25%). Even after Day 4, hand-authored will sit at ~33%, still 2× target.	Day 4 OpenRouter run lifts synthesis to ~24%. The hand-authored gap is documented in §3.3 and called out as a sensitivity caveat in the model card.

Risk / failure	Honest assessment	Mitigation (in scope)
Dev partition under-represented	30 tasks = 19.4% vs 30% target. Under-powered for early-stopping signal.	Day 4 synthesis splits 1/3 to dev (+20 tasks → 50 = 23.3%); Day 5 programmatic top-up adds 25 more → 75 = 30%.
Five categories train-blind	tone_drift, dual_control, gap_over_claiming, scheduling_edge, cost_pathology appear *only* in held_out (34 tasks total). Trained model has zero training signal on them.	Deliberate sealed-test diversity choice. Held_out delta on these 34 tasks reported separately as **generalization** not learned skill.
Single-rater IRA	κ = 0.95 is *intra*-rater (24h-blinded re-label by same author), not multi-annotator. A skeptical reviewer may treat this as soft.	No remediation possible in remaining time; documented honestly in `inter_rater_agreement.md` §Limitations.
C-005 (cost/lead $0.52) unverified	Marked unverified in `evidence_graph.json`; not cited in this report.	Remains unverified through submission; not load-bearing.
75 train tasks below LIMA threshold	LIMA showed 1,000 curated examples sufficient; we have 75 source tasks. Need augmentation to reach the 1,000–3,000 pair range required for stable LoRA.	Day 4 chat-template augmentation (§7.3) targets 75 → ~1,500 pairs via paraphrase + rejection sampling.

7.3 Path A-specific Day 4–7 plan

Path A is SFT with LoRA on Qwen 3.5 (0.8B or 2B, T4 16GB VRAM). The plan below is specific to Path A — it would not be the same for a Path B (DPO/preference-pair) or Path C (process supervision) trainee.

Day	Path A commitment	Path A papers cited	Output artifact
Day 4 AM	Format 75 train tasks into Qwen-3.5 ChatML chat template: system = phrasing-gate spec, user = `agent_prompt` + signals JSON, assistant = expected JSON output. Rejection-sampling quality filter: drop pairs where rubric dimensions disagree on PASS/FAIL with the deterministic evaluator.	LIMA (Zhou et al. 2023) — quality > quantity	`training/qwen_chat_format.jsonl`

Day	Path A commitment	Path A papers cited	Output artifact
Day 4 AM	**Augment 75 → ~1,500 pairs** via 20× paraphrase rotation: paraphrase `user_prompt` while holding `expected` JSON constant; reject paraphrases that change the inferred phrasing tier (LIMA-style filter). Target post-filter: 1,000–3,000 pairs.	LIMA (Zhou et al.); Tülu 3 (Lambert et al. 2024) — targeted SFT pipeline	`training/qwen_pairs.jsonl`, pair count logged
Day 4 PM	Colab T4 dummy LoRA (5 tasks, rank=16, alpha=32, q_proj+v_proj only, 16-bit). Push adapter to HF Hub to verify pipeline end-to-end before real run.	—	HF model card stub
Day 4 PM	OpenRouter synthesis run: `python generation_scripts/synthesis_generator.py --count 60`. Generator: Qwen3-80B. Judge: DeepSeek-chat-v3.2 (mean ≥ 3.5 admit). Generator ≠ Judge enforced. Split 40 train / 20 dev. **Eval-tier spend: ≤ $3.**	Magpie (Xu et al. 2024) — self-instruct synthesis	`train/TB-01xx.json`, `dev/TB-02xx.json`
Day 5 AM	Real LoRA training run on full 1,000–3,000 pair set. Hyperparameters: rank=16, alpha=32, q_proj+v_proj, 16-bit, lr=2e-4, batch=4, grad_accum=4, 3 epochs, **30-minute T4 wall-clock budget** per the brief. Convergence target: dev loss < 0.5.	Tülu 3 — hparam baseline	`training/loss_log.json`, `training/checkpoint/`
Day 5 PM	Delta A: trained LoRA vs Week 10 baseline on held_out. Paired bootstrap n=1000. Claude Sonnet 4.6 used as judge for soft dimensions only (stale_disclosed natural-language flag, abstention text quality). **Eval-tier spend: ≤ $4.**	—	`ablations/ablation_results.json`, fills C-006 in `evidence_graph.json`
Day 6 AM	Delta B: trained LoRA vs prompt-engineered Qwen 3.5 (same backbone, prompt-only). Tests whether SFT beats prompting. Negative result is publishable.	—	C-007 in `evidence_graph.json`
Day 6 PM	Model card (HF), blog post (HF Community), τ^2-Bench GitHub issue, HF dataset + adapter push (after staff sign-off).	—	HF artifacts

7.4 Eval-tier spend allocation against the $10 envelope

Phase	Item	Budget	Cumulative
Day 1–3	Bench authoring (zero API cost)	$0.00	$0.00
Day 4 PM	OpenRouter synthesis (Qwen3-80B gen + DeepSeek V3.2 judge, ~150 calls)	≤ $3.00	≤ $3.00
Day 5 PM	Claude Sonnet 4.6 held_out eval-tier judge calls (~50 soft-dim calls)	≤ $4.00	≤ $7.00
Day 6	Contingency (re-run, model-card eval, blog reproduction)	≤ $3.00	**≤ $10.00**

Every API call logged in `cost_log.md` within 24 hours. Hard cap enforced by deactivating OPENROUTER_API_KEY in `.env` when cumulative spend hits $9.50.

7.5 Kill criterion / pivot trigger (Day 5 training run)

The brief allots a 30-minute T4 window for the training run. The run is **killed and pivoted** if any of the following triggers fires:

Trigger	Threshold	Pivot action
Dev loss plateau	No improvement over 100 consecutive steps	Halve lr to 1e-4, restart from last checkpoint. If second 10-min run also plateaus, fall back to rank=8 / alpha=16 (smaller adapter).
Dev loss divergence	Rises > 20% over any 50-step window	Restart with weight_decay=0.01. If still divergent, drop epochs 3 → 1 and increase batch size to 8.
Mode collapse on dev	Generation samples show all outputs same `phrasing_tier` regardless of input (sampled at step 200, 400, 600)	Pivot to rank=32 / alpha=64 + oversample under-represented phrasing tiers in pair construction.
Wall-clock exhausted	Total Colab T4 time exceeds 30 min without convergence	**Hard pivot:** abandon Path A. Use prompt-engineered Qwen 3.5 (same backbone, no training) as the deliverable; report Delta B as the primary result; document Path A as an attempted-but-non-converged null in the model card. The bench, ablation harness, sealed held_out, and Delta B all still ship Saturday.

The wall-clock pivot is the explicit fall-back: **Path A non-convergence is a publishable null result, not a project failure.** The bench is the primary deliverable; the LoRA is the secondary deliverable; the bench ships regardless of training outcome.

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8. What Ships Saturday

- Full ~250-task dataset on HuggingFace (CC-BY-4.0); held_out released post-leaderboard.
- LoRA adapter on HuggingFace (Qwen 3.5, trained on ~1,500 augmented pairs from 115 source tasks) — *or*, if the kill-criterion fires, a documented Path A null and a prompt-engineered Qwen 3.5 baseline as the deliverable.

- Ablation results: Delta A (LoRA vs baseline) and Delta B (LoRA vs prompt) with paired-bootstrap confidence intervals, filling C-006 and C-007 in `evidence_graph.json`.
- 8 synthesis memos: LIMA, Magpie, Tülu 3 (done); Liu et al. COLM 2024, Gebru 2021, Pushkarna FAccT 2022, Chen et al. EMNLP 2025, Gu et al. 2024 (Days 5–6).
- Blog post (HF Community) and τ^2-Bench GitHub community engagement issue.

All numeric claims in this memo resolve to rows in `evidence_graph.json`. C-005 (cost/lead = $0.52) is marked unverified pending source confirmation and is not cited in this report.